19 CEI-56G-XSR-NRZ Extra Short Reach Interface

This clause details the requirements for the CEI-56G-XSR-NRZ extra short reach high speed electrical interface between nominal baud rates of 39.8 Gsym/s to 58.0 Gsym/s using NRZ coding. A compliant device must meet all of the requirements listed below. The electrical interface is based on high speed, low voltage DC-coupled logic with nominal differential impedance of 100 Ω and a channel with a nominal differential impedance of 92.5 Ω. Connections are point-to-point balanced differential pair and signalling is unidirectional.

The electrical IA is based on loss & jitter budgets and defines the characteristics required to communicate between a CEI-56G-XSR-NRZ driver and a CEI-56G-XSR-NRZ receiver using copper signal traces on a printed circuit board. The characteristic impedance of the signal traces is nominally 92.5 Ω differential. A 'length' is effectively defined in terms of its attenuation rather than physical length.

Extra short reach CEI-56G-XSR-NRZ devices from different manufacturers shall be interoperable.

19.1 Requirements

1. Support serial data rate from 39.8 Gsym/s to 58.0 Gsym/s. A CEI implementation complies to the specifications of this clause over the range of baud rates stated by the implementer within this range.
2. Capable of low bit error rate (10^{-15}, with a test requirement to verify 10^{-12}).
3. Capable of driving 0 – 50 mm of PCB trace.
4. Shall support DC-coupled operation.
5. Shall allow multi-lanes (1 to n).

19.2 General Requirements

19.2.1 Data Patterns

Please refer to Section 3.2.1.

19.2.2 Signal levels

The signal is a low swing DC coupled differential interface.

19.2.3 Signal Definitions

Please refer to Appendix 1.A.

A single clock signal is shared between the transmitting and receiving devices in the ingress direction and another single clock signal is shared between the transmitting and receiving devices in the egress direction, avoiding the need for a clock recovery circuit at the receiver.

19.2.4 Bit Error Ratio

Please refer to Section 3.2.3.

19.2.5 Ground Differences

As the driver and receiver are on the same PCB (with no intervening connectors), the ground difference is approximately 0 mV.

19.2.6 Cross Talk

Please refer to Section 3.2.5.

19.2.7 Channel Compliance

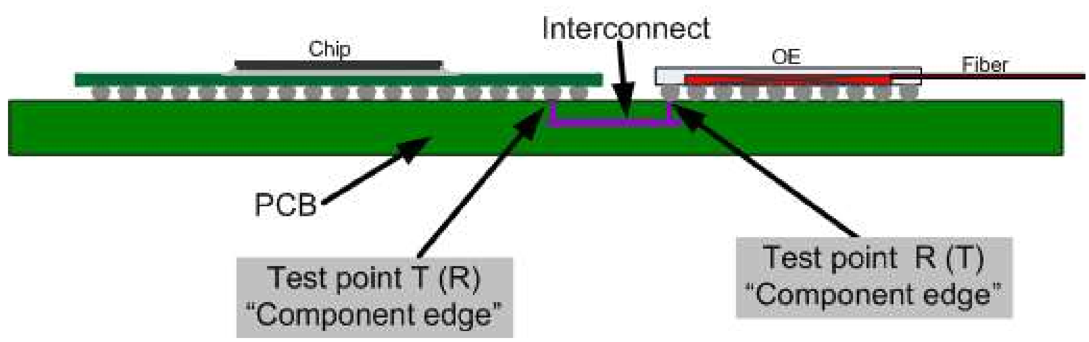
A forward channel and associated dominant crosstalk channels are deemed compliant if the channel characteristics conform to the requirements in this section.

19.2.7.1 Reference Model

The channel consists of PCB traces and any required vias. The reference PCB trace differential impedance is 92.5 Ω and all channel specifications are referenced to this impedance.

Figure 19-1 shows a diagram of the intended application.

Figure 19-1. CEI-56G-XSR-NRZ Reference Model



Several channel characteristics are parametrized according to Table 19-1 at these test points and are used for calculation of the channel parameters found in Table 19-2.

Table 19-1. Measured Channel Parameters

Symbol	Description
IL(*f*)	Differential insertion loss, -SDD21 magnitude (dB)
$RL_1(f)$	Differential input return loss, -SDD11 magnitude (dB)
$RL_2(f)$	Differential output return loss, -SDD22 magnitude (dB)
$NEXT_m(f)$	Differential near-end crosstalk loss (m^{th} aggressor), -SDD21 magnitude (dB)
$FEXT_n(f)$	Differential far-end crosstalk loss (n^{th} aggressor), -SDD21 magnitude (dB)

Table 19-2. Calculated Channel Parameters

Symbol	Description
$IL_{fitted}(f)$	Fitted insertion loss (dB)
ILD(*f*)	Insertion loss deviation (dB)
ICN(*f*)	Integrated crosstalk noise (mVRMS)
FOM_{ILD}	Channel Figure of merit - Weighted insertion loss deviation (dB)

19.2.7.2 Insertion Loss

Channel insertion losses, including PCB traces and connectors, shall comply with the limits specified by Equation (19-1), Equation (19-2) and plotted in Figure 19-2. Note that the variable f_b is the maximum baud rate to be supported by the channel under test (39.8 GHz < f_b < 58.0 GHz) and the measurement is to be made relative to the nominal impedance of 92.5 Ω.

Table 19-3. Channel Insertion Loss Frequency Range

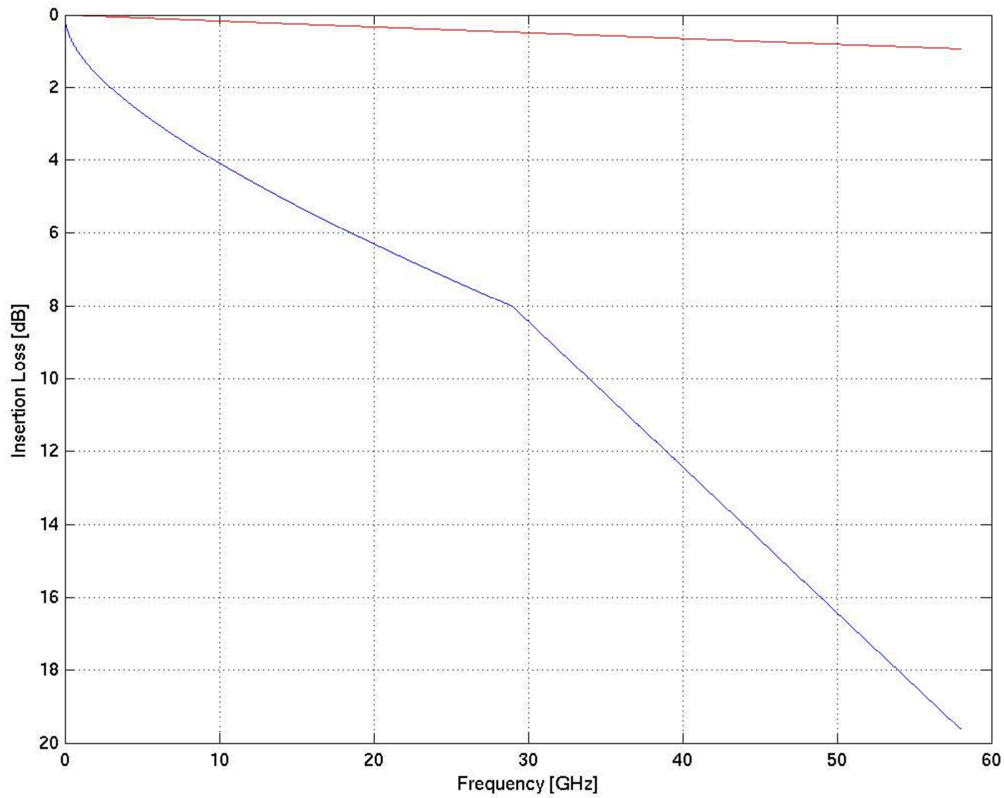
Parameter	Value	Units
f_{min}	50	MHz
f_{max}	f_b	GHz

$$IL_{max} = \left(\begin{array}{ll} 0.01 + 1.0\sqrt{\dfrac{f \times 58.0}{f_b}} + 0.09\dfrac{f \times 58.0}{f_b} & f_{min} \le f < \dfrac{f_b}{2} \\ -3.595 + 0.4\dfrac{f \times 58.0}{f_b} & \dfrac{f_b}{2} \le f \le f_b \end{array} \right) \quad (19\text{-}1)$$

$$IL_{min} = \left(0.009\sqrt{f} + 0.015f \quad f_{min} \le f < f_b \right) \quad (19\text{-}2)$$

Note: *f* in Equation (19-1) and Equation (19-2) is in GHz.

Figure 19-2. CEI-56G-XSR-NRZ normative channel insertion loss at 58.0 Gsym/s



19.2.7.3 Fitted Insertion Loss

For fitted insertion loss definitions, please refer to Section 12.2.1.1

The channel shall meet the insertion loss requirements defined in Table 19-4 and also meet the IL mask in Equation (19-1) and Equation (19-2). Note that the variable f_b is the maximum baud rate to be supported by the channel under test.

Table 19-4. Channel fitted insertion loss characteristics

Parameter	Units	Min Value	Max Value
Min frequency, f_{ILmin}	GHz	0.05	-
Max frequency, f_{ILmax}	GHz	-	fb
Fitted insertion loss at Nyquist	dB	-	8
Fitted insertion loss, a_0	dB	-1	2
Fitted insertion loss, a_1	dB	0	5
Fitted insertion loss, a_2	dB	0	16
Fitted insertion loss, a_4	dB	0	7.3

19.2.7.4 Insertion Loss Deviation (ILD)

The insertion loss deviation ILD is the difference between the measured insertion loss IL and the fitted insertion loss IL_{fitted} as defined in Equation (19-3) where f_{ILmin} and f_{ILmax} are given in Table 19-4.

$$ILD = IL - IL_{fitted} \qquad f_{ILmin} \le f \le (3/4) \times f_{ILmax} \tag{19-3}$$

The insertion loss deviation ILD shall be within the region defined by Equation (19-4).

$$-1.5 \le ILD \le 1.5 \tag{19-4}$$

A Figure Of Merit (FOM_{ILD}) for the channel is the weighted insertion loss deviation from f_{ILmin} to (3/4)*f_{ILmax}. FOM_{ILD} is calculated as indicated below. Note FOM_{ILD} is called ILD_{RMS} in OIF-CEI-03.0 clauses 10 & 11 and OIF-CEI-03.1 clauses 10, 11, 13 & 14.

Define the weight at each frequency *f* using Equation (19-5) below.

$$W(f) = \operatorname{sinc}^2(f/f_b)\left[\frac{1}{1+(f/f_t)^4}\right]\left[\frac{1}{1+(f/f_r)^8}\right] \tag{19-5}$$

Note that -3 dB transmit filter bandwidth f_t is inversely proportional to the minimum 20% to 80% rise and fall times T_*tr* and T_*tf*. The constant of proportionality is 0.2365 (i.e. min(T_*tr*, T_*tf*) x f_t = 0.2365, T_*tr* is in ns when f_t is in GHz). In addition, f_r is the -3 dB reference receiver bandwidth, which should be set at (3/4)f_b, where f_b is the maximum baud rate to be supported by the channel.

FOM_{ILD} is calculated using Equation (19-6) where N is the number of frequency points. The summation is done over the frequency range of ILD with *f* in GHz. FOM_{ILD} shall be less than 0.2 dB for compliant channels.

$$FOM_{ILD} = \sqrt{\frac{\sum W(f) \times ILD\langle f\rangle^2}{N}} \tag{19-6}$$

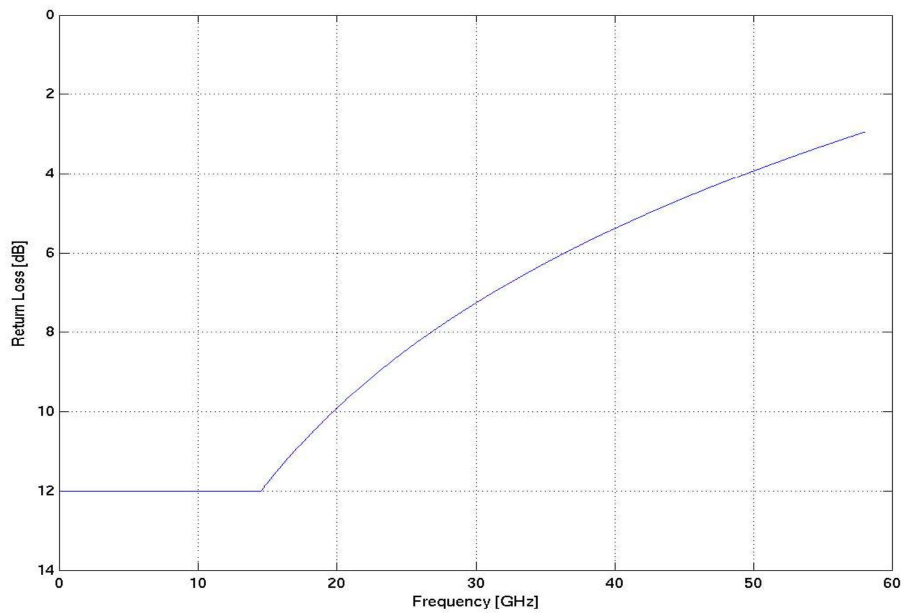
19.2.7.5 Channel Return Loss

Channel Return Loss shall be bounded by Equation (19-7) as shown in Figure 19-3 relative to the nominal impedance of 92.5 Ω.

$$\begin{aligned} &RL(f) > 12\ dB && f_{min} \le f \le \frac{f_b}{4} \\ &RL(f) > 12\ dB - 15\log_{10}\left(\frac{4f}{f_b}\right) && \frac{f_b}{4} < f \le f_b \end{aligned} \tag{19-7}$$

Note: f_{min} is as defined in Table 19-3

Figure 19-3. CEI-56G-XSR-NRZ normative channel return loss at 58.0 Gsym/s



19.2.7.6 Channel integrated crosstalk noise

Using the Integrated crosstalk noise method of Section 12.2.1.2 and the parameters of Table 19-5, the total integrated crosstalk noise for the channel shall be less than 2 mV_{RMS}.

Table 19-5. Channel integrated crosstalk aggressor parameters

Parameter	Symbol	Value	Units
Baud rate	f_b	max. Baud Rate supported by channel	Gsym/s
Near-end aggressor peak to peak differential output amplitude	A_{nt}	400	mVppd
Far-end aggressor peak to peak differential output amplitude	A_{ft}	400	mVppd
Near-end aggressor 20% to 80% rise and fall times	T_{nt}	4	ps
Far-end aggressor 20% to 80% rise and fall times	T_{ft}	4	ps

19.3 Electrical Characteristics

The electrical interface is based on high speed, low voltage DC-coupled logic with nominal differential impedance of 100 Ω and a channel with a nominal differential impedance of 92.5 Ω.

All devices shall work within the range 39.8 Gsym/s to 58.0 Gsym/s as specified for the device, with all ingress lanes synchronous to a common reference frequency having a stability of ± 100ppm from nominal and all egress lanes synchronous to a common reference frequency having a stability of ± 100ppm from nominal. The reference clocks of the ingress and egress directions are not necessarily synchronous to each other. Note that implementation of specific protocols will define the operating baud rate without affecting CEI compliance.

19.3.1 Reference Clock

Both ends of the link are to have a common clock frequency, which is set to be 1/64th of the baud rate. This clock could come from the same source or could be “forwarded” from the driver side to the receiver side. For details of applicable clock architecture options please refer to Annex 18.A.In a forwarded clock architecture the clock path needs to meet the same requirements as the data path. The electrical specifications at R_C are given in Table 19-6.

Table 19-6. Reference Clock Electrical Specification.

Characteristic	Symbol	Condition	MIN.	NOM.	MAX.	UNIT
Reference Clock Rate	Ref_Freq			$f_b/64$		GHz
Input Differential Voltage	Ref_Vdiff		240		900	mVppd
Input Single Ended Voltage	Ref_Vse		0.1		1.2	V
Differential Termination Resistance Mismatch	Ref_Rdm				5	%
Input Clock Rise and Fall Time (20% to 80%)	Ref_tr, Ref_tf		4		200	ps
Differential Input Return Loss	Ref_SDD11	at $f_b/128 \le f < f_b/4$ Note 1			-6	dB
Input Clock Duty Cycle	Ref_DC		40		60	%
High Frequency Uncorrelated Unbounded Gaussian Jitter	Ref_UUGJ_hf	Note 2			0.009	UI rms
Reference Clock Single Side Band Phase Noise	Ref_PN	@ 1kHz offset			-70	dBc/Hz
		@10kHz offset			-93	
		@100kHz offset			-113	
		@1MHz offset			-133	
		≥10MHz offset			-143	

NOTES:
1. Return loss is referenced to 100 Ohm
2. UUGJ measured using the methodology defined in Appendix 2.E.1 with a golden PLL cut-off frequency of fb/500, observing with a bandwidth of 43GHz

19.3.2 Transmitter Characteristics

The transmitter electrical specifications at compliance point T (see Figure 19-1) are given in Table 19-7. The transmitter shall satisfy jitter requirements specified in Table 19-8. Jitter is measured relative to the reference clock as timing source, using a golden clock multiplier as detailed in Annex 18.A.4, for a BER as specified in Section 3.2.3. It is assumed that the UBHPJ component of the transmitter jitter is not data-dependent jitter (DDJ) from the receiver view point, hence it cannot be equalized in the receiver. To attenuate noise and absorb even/odd mode reflections, the transmitter shall satisfy the Common Mode Output Return Loss requirement of Table 19-7. The waveform is observed through a fourth-order Bessel-Thomson response with a bandwidth of 43 GHz using a PRBS31 pattern.

The link budget in this document assumes no Tx emphasis.

Table 19-7. Transmitter Electrical Output Specification.

Characteristic	Symbol	Condition	MIN.	TYP.	MAX.	UNIT
Baud Rate	T_Baud		39.8		58.0	Gsym/s
Output Differential Voltage	T_Vdiff		250		400	mVppd
Single Ended Transmitter Output Voltage Note 1	T_Vse		0.1		1.2	V
Differential Termination Resistance Mismatch	T_Rdm				5	%
Output Rise and Fall Time (20% to 80%)	T_tr, T_tf		4			ps
Differential Output Return Loss	T_SDD22	See 19.3.2.2 Note 2				dB
Common mode Output Return Loss	T_SCC22	200MHz to f_b/2			-6	dB
Transmitter Common Mode Noise	T_Ncm				15	mVrms

NOTES:
1. DC Coupling compliance is mandatory.
2. Return loss is referenced to 100 Ohm

Table 19-8. Transmitter Output Jitter Specification

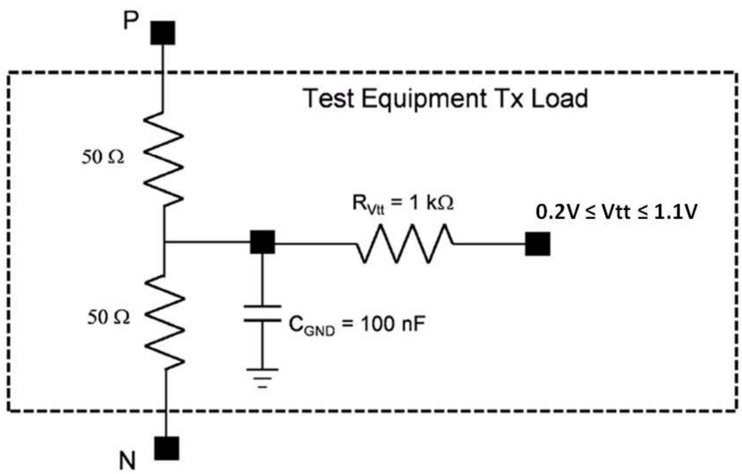
Characteristic	Symbol	Condition	MIN.	TYP.	MAX.	UNIT
Uncorrelated Unbounded Gaussian Jitter	T_UUGJ				0.15	UIpp
Uncorrelated Bounded High Probability Jitter	T_UBHPJ				0.15	UIpp
Even Odd jitter (component of UBHPJ)	T_EOJ	Note 2			0.035	UIpp
Total Jitter	T_TJ	Note 1			0.28	UI
Eye Mask	T_X1	See 19.3.2.4			0.14	UI
Eye Mask	T_X2	See 19.3.2.4			0.4	UI
Eye Mask	T_Y1	See 19.3.2.4	125			mV
Eye Mask	T_Y2	See 19.3.2.4			200	mV

NOTES:
1. T_TJ includes all of the jitter components.
2. Included in T_UBHPJ. Even-odd jitter is defined in Table 1-3.

19.3.2.1 Transmitter Amplitude and Swing

Transmitter differential output amplitude shall be able to drive between 250 and 400 mVppd. Further the single-ended voltage must be between 0.1 and 1.2 V. Figure 19-4 shows the transmitter test load configuration.

Figure 19-4. Transmitter Amplitude Test Load



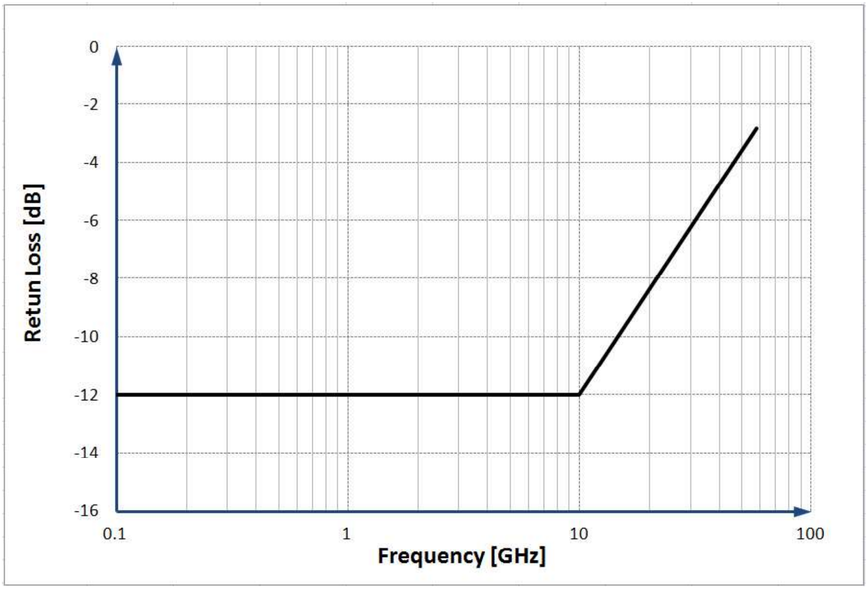
19.3.2.2 Transmitter Resistance and Return Loss

Please refer to Section 3.2.10 with the following parameters in Table 19-9 and as illustrated in Figure 19-5.

Table 19-9. Driver Return Loss Parameters

Parameter	Value	Units
A0	-12	dB
f0	100	MHz
f1	0.1714 x T_Baud	Hz
f2	T_Baud	Hz
Slope	12.0	dB/dec

Figure 19-5. Illustration of Return Loss for T_Baud = 58.0 Gsym/s



19.3.2.3 Transmitter Lane-to-Lane Skew

Please refer to Section 3.2.7.

19.3.2.4 Transmitter Template and Jitter

When provided with the reference clock meeting the specifications in Section 19.3.1 as a timing source, using a golden clock multiplier as detailed in Annex 18.A.4, for a BER as per Section 3.2.3, the transmitter shall satisfy the eye template and jitter requirements as given in Table 19-8 and Figure 1-4. The measurement of jitter and eye diagram is to be relative to the “forwarded” clock if the transmitter provides one. If it does not provide a “forwarded” clock then the measurement is to be relative to the reference clock provided to it.

19.3.3 Receiver Characteristics

A compliant receiver shall operate at the specified BER with the worst case combination of a compliant transmitter and a compliant channel.

Receiver electrical specifications are given in Table 19-10 and measured at compliance point R. To dampen noise sources and absorption of both even and odd mode reflections, the receiver shall satisfy the Common Mode Input Return Loss requirement of Table 19-10.

Table 19-10. Receiver Electrical Input Specification

Characteristic	Symbol	Condition	MIN.	TYP.	MAX.	UNIT
Baud Rate	R_Baud		39.8		58.0	Gsym/s
Input Differential Voltage	R_Vdiff	Note 1			400	mVppd
Receiver Common Mode Noise Tolerance	R_Ncm				25	mVrms
Differential Termination Resistance Mismatch	R_Rm				5	%
Differential Input Return Loss	R_SDD11	See 19.3.3.2 Note 2				dB
Common mode Return Loss	R_SCC11	200 MHz to10 GHz			-6	dB
		10 GHz to f_b			-4	dB
Rx Input Single Ended Voltage Range	R_V_{inSE}		0.1		1.2	V

NOTES:
1. The receiver shall have a differential input voltage range sufficient to accept a signal produced at point R by the combined transmitter and channel. The channel response shall include the worst case effects of the return losses at the transmitter and receiver
2. Return loss is referenced to 100 Ohm

When provided with the reference clock meeting the specifications in Section 19.3.1 as a timing source, the receiver shall tolerate the sum of these jitter contributions and meet the BER as per Section 3.2.3: the total transmitter jitter from Table 19-8 and the effects of a channel compliant to the Channel Characteristics (Section 19.2.7).

19.3.3.1 Reference Receiver Input Signals

The receiver shall accept differential input signal amplitudes produced by a compliant transmitter connected with the minimum attenuation specified in Equation (19-2) to the receiver. This may be larger than the 400 mVppd maximum of the transmitter due to output/input impedances and reflections.

The minimum input amplitude is defined by the minimum transmitter amplitude, the actual receiver input impedance and the loss of the actual PCB. Note that the minimum transmitter amplitude is defined using a well controlled load impedance; however the real receiver's impedance may differ, which can leave the receiver input signal smaller than expected. Additionally the real receiver may be affected by environmental noise.

19.3.3.2 Receiver Return Loss

Please refer to Section 3.2.10 with the following parameters in Table 19-11 and as illustrated in Figure 19-5.

Table 19-11. Receiver Input Return Loss Parameters

Parameter	Value	Units
A0	-12	dB
f0	100	MHz
f1	0.1714 x R_Baud	Hz
f2	R_Baud	Hz
Slope	12.0	dB/dec

19.3.3.3 Input Lane-to-Lane Skew

Please refer to Section 3.2.8.

20 CEI-56G-XSR-PAM4 Extra Short Reach Interface

This clause details the requirements for the CEI-56G-XSR-PAM4 extra short reach high speed electrical interface between nominal baud rates of 18.0 Gsym/s and 29.0 Gsym/s using PAM-4 coding. A compliant device shall meet all of the requirements listed below. The electrical interface is based on high speed, low voltage DC-coupled logic with nominal differential impedance of 100 Ω and a channel with nominal differential impedance of 92.5 Ω. Connections are point-to-point balanced differential pairs and signaling is unidirectional.

The electrical IA is based on loss and jitter budgets and defines the characteristics required to communicate between a CEI-56G-XSR-PAM4 transmitter and a CEI-56G-XSR-PAM4 receiver using copper signal traces on a printed circuit board. Refer to Section 20.2.7 for channel requirements.

Extra short reach CEI-56G-XSR-PAM4 devices from different manufacturers shall be interoperable.

20.1 Requirements

1. Support serial baud rates within the range from 18.0 Gsym/s to 29.0 Gsym/s. A CEI implementation complies to the specifications of this clause over the range of baud rates stated for the implementation within this range.
2. Capable of achieving a raw Bit Error Ratio performance of 10^{-9} or better per lane. FEC is assumed to be used in the system to achieve corrected BER of 10^{-15} or better per lane. The baud rate includes the overhead required for FEC. The definition of FEC is outside the scope of this IA (see Appendix 16.D).
3. Capable of driving 0 - 50 mm of PCB trace.
4. Shall support DC-coupled operation.
5. Shall allow multi-lanes (1 to n).

20.2 General Requirements

20.2.1 Data Patterns

See 16.C.5.

20.2.2 Signal levels

The signal is a low swing DC-coupled differential interface.